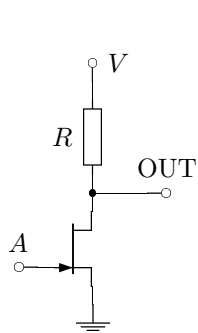


CIRC—例とコード



1. NOT

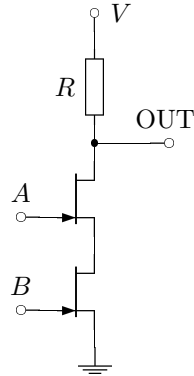


図 2. NAND

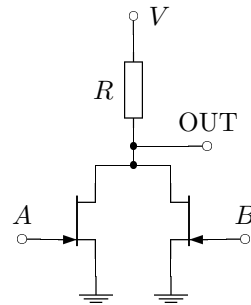


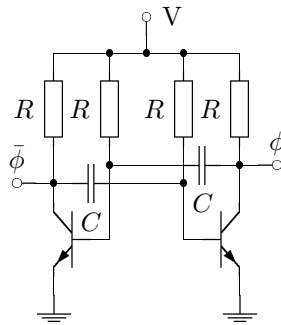
図 3. NOR

```
\begin{circuit}0
%NOT
\nfet1 {?} G 1 % transistor
\frompin nfet1S
\ - 1 d
\gnd1
\frompin nfet1D
\ - 1 u
\ .1
\frompin .1
\ - 1 u
\nl\R1 {$R$} u
\ - 1 u
\cc\connection1 {$V$} c u
\frompin nfet1G
\ - 1 l
\cc\connection2 {$A$} c l
\frompin .1
\ - 3 r
\cc\connection3 {OUT} c r
\end{circuit}
\begin{circuit}0
%NAND
\nfet1 {?} G 1
\frompin nfet1S
\ - 1 d
\nfet2 {?} D 1
\frompin nfet2S
\ - 1 d
\gnd1
\frompin nfet1D
\ - 1 u
\ .1
```

```

\frompin .1
\ - 1 u
\nl\R1 {\R$} u
\ - 1 u
\cc\connection1 {\V$} c u
\frompin nfet1G
\ - 1 l
\cc\connection2 {\A$} c l
\frompin nfet2G
\ - 1 l
\cc\connection3 {\B$} c l
\frompin .1
\ - 3 r
\cc\connection4 {OUT} c r
\end{circuit}
\begin{circuit}0
%NOR
\nfet1 {?} G l % transistor
\frompin nfet1S
\ - 1 d
\gnd1
\frompin nfet1D
\ - 1 u
\ - 2 r
\ .1
\frompin .1
\ - 1 u
\ .2
\frompin .2
\ - 1 u
\nl\R1 {\R$} u
\ - 1 u
\cc\connection1 {\V$} c u
\frompin .1
\ - 2 r
\ - 1 d
\nfet2 {} D r
\frompin nfet2S
\ - 1 d
\gnd2
\frompin nfet1G
\ - 1 l
\cc\connection2 {\A$} c l
\frompin nfet2G
\ - 1 r
\cc\connection3 {\B$} c r
\frompin .2
\ - 3 r

```



```
\cc\connection4 {OUT} c r
\end{circuit}
```

```
\begin{circuit}0
  \npn1 {?} B r
  \frompin npn1E
  \- 1 d
  \gnd1
  \frompin npn1B
  \- 1 r
  \- 4 u
  \.1
  \- 1 u
  \nl\R1 {$R$} u
  \- 1 u
  \.2
  \frompin npn1C
  \- 1 u
  \.3
  \- 2 u
  \nl\R2 {$R$} u
  \frompin R2t
  \vtopin .2
  \htopin .2
  \frompin .2
  \- 2 r
  \.4
  \- 2 r
  \.5
  \- 1 d
  \nl\R3 {$R$} d
  \- 2 d
  \.6
  \- 3 d
  \- 1 r
  \npn2 {?} B 1
  \frompin npn2E
  \- 1 d
```

```

\gnd2
\frompin npn2C
\~ 2 u
\ .7
\~ 1 u
\nl\R4 {\R$} u
\frompin R4t
\vtopin .5
\htopin .5
\frompin .3
\~ 1 r
\nl\C1 {\C$} r
\frompin C1r
\htopin .6
\frompin .7
\~ 1 l
\nl\C2 {\C$} l
\htopin .1
\frompin .7
\~ 1 r
\cc\connection1 {\$\phi$} c r
\frompin .4
\~ 1 u
\cc\connection1 {V} c u
\frompin .3
\~ 1 l
\cc\connection1 {\$\bar{\phi}$} c l
\end{circuit}

```